Jinming Ren

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Education

University of Electronic Science and Technology of China (UESTC) Sept 2022 — Present

University of Glasgow, Dual Degree Program Sept 2022 — Present

- **Major**: Electrical & Communication Engineering BEng; GPA: 3.87/4.0, Ranking: 2/164 (Top 1.2%)
- **Relevant Coursework**: Information Theory, Artificial Intelligence and Machine Learning, Stochastic Processes, Digital Circuit Design, etc.
- **Online Course**: Abstract Algebra, Complex Analysis, Differential Geometry, Convex Optimization, etc.

Research & Projects

System-level Co-Design of RISCV Accelerators for TinyML at the Edge Sept 2025 — April 2026

Research Assistant, Prof. Yun Li, UESTC

- Engineered a standalone Neural Processing Unit (NPU) for real-time YOLOv8n edge inference on an Artix-7 FPGA, bypassing soft-core processors via a custom RISC-V instruction extension (Xnpu).
- Architected an end-to-end Python ML compiler for automated INT16 quantization and memory-aware instruction scheduling, preserving accuracy within 0.3% mAP of the PyTorch FP32 baseline.
- Designed parameterized RTL operators featuring a 3x3 systolic MAC grid and fully hardware-accelerated post-processing (DFL, NMS), achieving 288 MACs/cycle and 23.4 GOPS peak throughput.
- Integrated asynchronous camera/UDP video pipelines and AXI4 DDR3L memory multiplexing, fully verified via `Cocotb`, `Bazel`, and `Icarus Verilog`.

YOPO: You Only Pick Once — Light Object Tracking Algorithm Sept 2025

- Developed a lightweight object tracking algorithm that requires only one initial selection, successfully mitigate the intense computation of DNN forward propagation on every frame.
- Utilized NCC-based matching, adaptive kernel updating, capable of tracking objects with gradual color and size changes.

Design and Visualization of a Complete Single-cycle RV32I CPU Core Jan 2025 — Mar 2025

- Designed a single-core, single-cycle RISCV 32-bit CPU from scratch in `Verilog` for RTL simulation and in `Digital` Software for working principle visualization, open-sourced on Github.
- Built a complete datapath including PC, fetcher, decoder, register file, ALU, LRU-based L1 cache, etc., compatible with basic peripherals: GPIOs, IIC, UART, etc.
- Implemented a boot program in RISCV assembly, basic delay and GPIO libraries in `C`. Compiled and simulated using RISCV GNU toolchain.

CNN/LSTM for Embedded Systems Feb 2024 — May 2024

- Designed and Integrated CNN and LSTM models into STM32 MCU for end-to-end patient fall detection of accuracy 95%, temperature monitoring and real-time data visualization.
- Manually collected and labeled time-series 3D acceleration dataset. Trained models on Linux, then hard-coded and accelerated them in `C++` on MbedOS for real-time inference.

Relevant Skills

IT Skills	Latex, Quarto Markdown, Linux, Manim, Github.
Programming	`C/C++`, `Python`, `RISCV Assembly`, `Verilog`, `Makefile`, `Bazel`, `Chisel`, `Matlab`.
Language	Native Chinese, Fluent English (IELTS 7).

Awards

Top Academic Scholarship of UESTC (Top 5%) Dec 2023, Dec 2024

China National Scholarship (Top 0.2%) Dec 2024

First Prize: 7th National College Art Exhibition and Performance (Violin section) Sept 2024

Last updated: May 12, 2026